

Instructional Animation, Segmentation and User Control Strategies

Anuar Hassan¹, Ahmad Zamzuri Mohamad Ali², Mohd Najib Hamdan³
Multimedia Department,
Universiti Pendidikan Sultan Idris
Tanjong Malim, Perak, Malaysia
anuar5225@gmail.com¹, zamzuri@fskik.upsi.edu.my², najib.kp@yahoo.com³

Abstract- The aim of this study is to evaluate the effect of segmented animation with various user control strategies on students' achievement. The research design was in a quasi-experimental form and the research data were analyzed using a quantitative approach. Sample sizes of 265 semester two students pursuing the Diploma in Networking System from five polytechnics were involved in the study. Overall, the ANCOVA statistical test was used to answer the research questions. The research findings revealed significant differences in students' achievement among groups who had received different control strategies of learning. The group who studied the Random User Control (KPR) strategy obtained highest mean score and the group who studied the Without User Control (TK) strategy obtained lowest mean score. Therefore, this study recommended the Random User Control (KPR) as the best control strategy for designing the segmented animation courseware.

Keywords—animation; segmentation; user control; program control; control strategy.

I. INTRODUCTION

Animation seen to has many advantages and it is supported by theories, however the findings on the effectiveness of animation in learning are still inconsistent [1]. The main cause of the inconsistency in findings is due to the lack of aspect of design and differences in student learning features, such as visualization ability, learning style, motivation, prior knowledge, age and gender [2].

From the design's perspective, animation features that change over time will make the students' feel confused and disordered when the animation is displayed. This is because they are not able to process the relentless sequence of information in the memory system due to the capacity of information received at one time is very limited [3]. It is also potentially mislead the students focus and attention towards important information conveyed in an animated presentation [4]. Therefore, the design of animation that does not provide appropriate time for students to concentrate on the information presented, is one of the factors that contribute to the failure of effective animation learning [5]. Without giving an ample time, students will failed to build a perfect mental model in their memory structures [6]. Most instructional animation applications might be less effective because the designer failed to take

into account the challenges of processing this information in their design [7].

The speed and complex visual factors are the probable causes that make content understanding among students obscure on the subjects being shown [8]. This is because they fail to extract and process continuously relentless information at a go [9]. Therefore an animation design that does not take these into consideration would distort students' focus and attention thus failing the role of animation as teaching aids [2].

A previous research done by [10] on a group of 143 teacher trainees has successfully proved the effectiveness of this segmental animation in teaching. In the study, they were taught seven basic teaching skills by an expert in the form of video animation. Based on the post and consistency tests results, the group tested using segmented animation was more prominent than to those using uncontrolled continuous animation methods. This is due to the fact that the rate of information received by the working memory is sufficient and the cognitive load is at the minimum level. Nevertheless, it was not stated clearly anywhere in the study about the form of control used in the animation.

One of the solutions in addressing this problem is animations are segmented with a control strategy to monitor the displayed segments, compared to passively look at the displays [8]. The findings also showed that segmented animations will allocate an appropriate time for students to interpret and understand the animations effectively [9]. However, to ensure the effectiveness of learning, the correct selection of segmented animation control must be put into consideration [2]. There are many previous studies related to animation in learning process, but very few studies have focused on segmented animation control strategies. Generally, there are two control strategies that can be used in animation segmented display which is user control and program control [2].

II. METHODOLOGY

A. Objective

This study aims to examine the effects of five segmented animation featured by various user control strategy, namely Linear User Control (KPL), Random User Control (KPR),

Free User Control (KB), Program Controlled (KP) and Without User Control (TK) to test the achievement of post test of polytechnic students. All control strategies were randomly assigned to each selected polytechnics.

A. Instructional Media

Segmented animation courseware prototype development is based and according to the principles of design of multimedia instruction DDD-E model developed by [7]. Briefly, the difference between these five segmented animations is the aspect of the user's control strategy that is used as follows:

1) Linear User Control (KPL)

Fig.1 shows the example of the strategy of KPL. Animation is divided into several meaningful segments. Students can advance or reverse from one segment to another according to their needs by pressing the available forward and reverse buttons. Students can only advance and retreat in a linear sequence from one segment to another.



Fig. 1. Example of Linear User Control Strategy (KPL)

2) Random User Control (KPR)

Animation is divided into several meaningful segments. Students can watch any animated segments according to their needs by pressing the relevant segments and it is randomly selected according to student needs. Fig.2 shows the sample navigation control of KPR.



Fig. 2. Example of Random User Control Strategy (KPR)

3) Program Controlled (KP)

Animation is divided into several meaningful segments. The Movements from one segment to another is determined by the animation development. The segmented animation display is entirely controlled by computer. Fig. 3 show the position of navigation icon for KP control strategy. Overall segmented animation is displayed linearly with intervals between each segment. Students can repeatedly playback the segmented animation by pressing the reset button.



Fig. 3. Example of Program Controlled Strategy (KP)

4) Free User Control (KB)

Fig.4 shows the example of the strategy of Free User Control. Animation will be shown continuously from beginning to end when the student presses the start button.

Students can stop the animation at any part by pressing the stop button and playback by pressing the hold button.



Fig. 4. Example of Open User Control Strategy (KB)

5) Without User Control (TK)

Animation will be shown continuously from beginning to end when the student presses the start button. Students can repeat the animation displayed by pressing the reset button. Fig.5 shows the sample navigation control of Free User Control.



Fig. 5. Example of Without User Control Strategy (TK)

B. Research sample

The sample for this study consists of semester two students in Network Systems Diploma program from five polytechnics and selected through multistage sampling. Two classes were selected from each polytechnic and they were determined by the Department of Information Technology and Computer respectively. Accordingly, a total of 265 students were taken as the sample.

Upon completion of the pre test, students begin their learning session for 20 minutes in the laboratory. Post test was performed once students complete the learning activities. Time allocated for pre and post test is 40 minutes. The content of question for both pre and post test are similar but different in aspect of arrangement. All the questions develop base on the standard of Table of Specification for Test (TOS) which is controlled and monitored by Polytechnics department.

C. Research Procedure

This is a quasi-experimental factorial design with pre and post study conducted in five polytechnics under the Ministry of Education (MOE). The five selected polytechnics are offering diploma in network system courses. All of these polytechnic are among conventional polytechnics. Two lecturers were appointed to supervise the pre and post test. Briefing about the courseware and regulatory oversight were given to assistant lecturer before the study started.

D. Research Instrument

The instruments used in this study were pre-test and post-test applied in conventional way, i.e. using paper and pen. These tests are intended to test and determine the level of understanding of abstract concepts contained in the Network Fundamentals courseware modules.

III. DATA ANALYSIS

Overall, ANCOVA analysis was used to test whether there is any significant difference to the performance of the post-test between students who received various control strategies of segmented animation. Meanwhile, the covariate used to answer the question of this study was pretest. Before that, several conditions have to be fulfilled to ensure that research data can be used to answer the question in this study using ANCOVA analysis. The findings of the analysis is presented in Table 1.0, after adjustment, post-test scores showed significant differences in terms of student achievement in post-test control strategies based on the segmented animation display KPL, KPR, KB, KP dan TK, $F(4, 259) = 5.99$, $p < 0.05$, *partial eta Square* = 0.09, with effect of sizes ranging between medium and high according to the Cohen guidelines 1998.

Meanwhile, the details of the analysis results shows that Random User Control (KPR) strategy obtain the best performance after post-test scores adjustment, ($M=67.75$, $SE= 1.56$) followed by KPL ($M= 67.33$, $SE=1.67$), KP ($M=61.76$, $SE=1.57$) dan KB($M=60.71$, $SE = 1.57$) and TK ($M = 58.86$, $SE = 1.67$).

TABLE I. SUMMARY OF TESTS BETWEEN- SUBJECTS EFFECTS

Control Strategy	N	Unadjusted Mean		Adjusted Mean	
		Mean	Std. Error	Mean	Std. Error
KPR	57	67.88	1.82	67.75	1.56
KPL	49	67.28	1.68	67.33	1.67
KP	55	61.75	1.59	61.76	1.57
KB	53	60.58	1.62	60.71	1.60
TK	51	58.33	1.79	58.86	1.67

IV. DISCUSSION

The overall results of the study showed significant differences exist in terms of post-test performance between students who use segmented animation with different control strategies. Based on the findings in pair comparison test, there is no significant difference to the performance of groups of students who follow the strategy of KPR, KPL and KP. While the performance of groups of students who follow the strategy of KPR, KPL and KP are much better than the strategy of KB and TK. However, statistically, there was no significant difference between KP, KB and TK strategies. This finding indicates that the segmented animation accompanied with user control, either linear or random is more helpful than learning continuous animation display. This finding is consistent with the previous studies

done by [8], which concluded that the segmented animation with linear manual control is better than continuous exposure animation. It was inline with the findings of [9] which had been stated that the segmented animation with linear user control strategies, produce better performance than independent control strategies and without the control strategy. Meanwhile, previous research delivered by [4] also proved that the effectiveness of segmented animation with linear control strategy compared to the continuous animation display without control. Moreover, according to [2], it was concluded that segmented animation with linear user's control strategy is better in improving student achievement than continuous animation exposure without control. Although these studies are limited to comparing the segmented animation with continuous animation display, but it is clearly shows that the segmented animation viewed as a linear with any of user control is still better than the continuous animation display.

Segmented Animation and displayed in a linear is seen to have advantages from the point of allowing students to allocate an appropriate time to extract and interpret useful information throughout the learning process, especially in the learning of abstract topics [2]. This, potentially overcome the problem of missing of information involved in the mental model formation for effective learning. Segmented animation also seen helping to promote active and passive processing in the student's memory structure. Active processing occurs during the presentation of animation while in a passive processing occurs when an animation stop for a while before move on to next segment. The combinations of dynamic and static display potentially overcome the problem that may exist when an item is purely viewed dynamically, as outlined by daylight.

In contrast with continuous display of such in KB and TK strategies, stream of information that enters memory structure would allows students to drop useful information [11]. This does not help in the process of building the perfect mental model related to the topic being studied and likely contributed to the failure of effective learning in a short period. Although students are given the freedom to stop and play the animation, this feature does not provide effective learning if the segment is determined by the courseware designer. Students may not know where the significant stopping point is if it is an abstract topic. Further studies that look at the interaction of students to remote control features would be able to answer this issue. Meanwhile, the TK strategy seems not to be a good option for studying abstract topics. This strategy does not able to provide appropriate time to the student and causes students focus and attention affected and the role as a teaching aid potentially fail [12].

V. CONCLUSION

Based on the findings obtained from the study, a number of findings and conclusions have been made. Generally, the design of Random User Control Strategy (KPR) and Linear User Control Strategy (KPL) has a significant impact on the performance of the post-test and it is better than Program Linear Controlled (KP), Free User Control (KB) and Without User Control strategy (TK). Accordingly, the animations displayed should be divided into small meaningful segments using user control strategy, either at random user controlled strategy or linear user controlled strategy (KPL) for a better students' achievement.

REFERENCES

- [1] H. Lin, and F.M. Dwyer, "The effect of static and animated visualization: A perspective of instructional effectiveness and efficiency". Educational Technology Research and Development, 2010, vol. 58, no. 2, pp. 155-174.
- [2] I. A. Spanjers, T. van Gog and J. J. van Merriënboer, "A theoretical analysis of how segmentation of dynamic visualizations optimizes students' learning". Educational Psychology Review, 2010, vol. 22, no. 4, pp. 411-423.
- [3] A. Wong, W. Leahy, N. Marcus and J. Sweller, "Cognitive load theory, the transient information effect and e-learning". Learning and Instruction, 2012, vol. 22, no. 6, pp. 449-457.
- [4] F. Wilson and F. M. Dwyer, "Effect of time and level of visual enhancements in facilitating student achievement of different educational objectives". International Journal of Instructional Media, 2001, vol. 28, no. 2, pp. 159-167.
- [5] J. Torres and F. Dwyer, "The effect of time in instructional effectiveness of varied enhancement strategies". International Journal of Instructional Media, 1999, vol. 8, no. 4, pp. 2-8.
- [6] B. Mireille and R. Cyrill, "Building dynamic mental models from animation: Effect of user control on exploration behaviours". Proceedings of the 12th EARLI biennial Conference on Research and Learning and Instruction, 2007, pp. 188-192. [Online]. Available: http://tecfa.unige.ch/perso/mireille/papers/Betrancourt_RebetzEARLI07.pdf.
- [7] R. K. Lowe, "Interrogation of a dynamic visualization during learning". Learning and Instruction, 2004, vol. 14, no.3, pp. 257-274.
- [8] R. Moreno, "Optimising learning from animations by minimizing cognitive load: Cognitive and affective consequences of signalling and segmentation methods". Applied Cognitive Psychology, 2007, vol. 21, no. 6, pp. 765-781.
- [9] A.Z.M. Ali, "Effects of Segmented-Animation in Projected Presentation Condition". Educational Technology & Society, 2013, vol.16, no. 3, pp. 234-245.
- [10] R. Moreno, "Optimising learning from animations by minimizing cognitive load: Cognitive and affective consequences of signalling and segmentation methods". Applied Cognitive Psychology, 2007, vol. 21, no. 6, pp. 765-781.
- [11] M. Betrancourt. "The animation and interactivity principles in multimedia learning", R. E. Mayer, Ed. The Cambridge handbook of multimedia learning New York, NY: Cambridge University Press, 2005, pp. 287-29.
- [12] A.Z.M. Ali, "Kesan koswer strategi animasi pelbagai kawalan pengguna terhadap prestasi kognitif dan ketekalan pelajar". PhD Thesis, Universiti Sains Malaysia, Malaysia, 2007.